SEEING CHANGE FROM ORBIT · MEETING MINUTES

Meeting 3: Data Preparation And Fusion Planning

Date and time	Mon 21 Sep 2026, 1:30–2:30 pm
Type	Supervisor
Location	In person (Dr. Panos's office, D04208)
Attendees	Dr. Panos, Zain, Tehsin

Agenda

1. Team plan and kickoff document (sent 20 Sep)
2. Live demo of data, files, tools and code
3. Dates and labels
4. Image registration
5. Fusion of observations over time
6. Validation, existing tools and external data

Member updates

Who	Update
Tehsin	Added the project tracks, SDP I plan and data documentation to GitHub (19 Sep); presented the live data, files, tools and code to Dr. Panos with Zain
Zain	Sent the team plan and agenda to Dr. Panos (20 Sep); presented the live demo with Tehsin
Zain, Tehsin	Briefed the members who couldn't attend (Ayesha, Zayed, Khaled)

Discussion

1. Team plan

- Presented the team plan: five research jobs to 11 Oct, two people per job (see team documents).

2. Data demo and data understanding

- Current Earth Engine / GIQ pulls had only 3 bands (RGB). Sentinel-2 has 13 bands; add NIR (B8) and SWIR (B11, B12). Hyperspectral not needed.
- Before processing, document what each file contains: source and product, size, resolution, bands and their meaning, data type, bit depth, documented value range. Read it from the source docs, not from displayed values. Keep optical bands and SAR distinct.

3. Dates and labels

- We proposed 2018 vs 2025 (long interval, satellite availability) and GMW maps as weak training labels, with selected areas hand-checked for testing.
- Dr. Panos: acceptable in principle; sampling and labelling protocol still to be defined.

4. Image registration

- Align with a geometric transform (translation, rotation, scale) on landmarks unaffected by tide, e.g. the hotel near Eastern Mangroves. A moving water boundary is not a landmark. This is separate from tide matching.

5. Fusion over time

- Central research question: how long can a mangrove area be treated as unchanged?
- Matching tides alone is not enough if images are far enough apart for real change.
- Two intervals: 2018–2025 = long-term change; fusion window = short no-change span.
- Method: one sensor first; from a start observation step forward until change is first detected; repeat over start dates, seasons and sites; use the shortest interval as a conservative window; within it, combine sensors at comparable tides. No window length chosen yet.
- eo-tides discussed as a way to select comparable tides; no workflow adopted.

6. Validation, existing tools and external data

- Check the workflow by comparing long-term gain/loss with GMW before estimating short windows. Tolerance not set.
- GMW is yearly. First check whether GMW, another dataset or an existing tool gives finer temporal data. If not, a registration and change tool becomes an early project component.
- ADIA Lab (climate science contact) could provide data or advice. Dr. Panos is open to it but wants free sources checked first. We offered to draft an email for his review.

Decisions

- Imagery must be multispectral (add B8, B11, B12); hyperspectral not needed.
- First deliverable is a data dictionary per source, including GIQ.
- 2018 vs 2025 with GMW weak labels and hand-checked test squares: approved in principle.
- Register on tide-stable landmarks with a geometric transform.
- Measure a fusion window before fusing; don't fuse on tide alone.
- Order of work: one sensor → registration + change → validate against GMW → fusion window → multi-sensor fusion.
- Search for finer-than-yearly data before building anything.
- Meetings become 1 hour, in person; deliverables are posted to the Teams group.

Action items

Who	What	Due
To assign	Data dictionary (all imagery incl. GIQ)	Not set
To assign	Export Sentinel-2 with all spectral bands	Not set
To assign	Check GMW / alternatives for monthly or finer data or change info	Not set
To assign	Reusable registration and change method for one sensor	Not set
To assign	Fusion approaches: options, pros and cons	Not set
To assign	Plan the GMW comparison: region and period	Not set
To assign	Plan the fusion-window study: sites, start dates, how change is detected	Not set
To assign	Draft ADIA Lab inquiry for Dr. Panos to review	Not set
Team	Full who-does-what plan and what SDP I delivers	Next meeting

Open questions: available acquisition frequency and exact products; separating real change from tide, misalignment and noise; fusion-window sampling; validation reference, tolerance and role of eo-tides.

Next meeting

Mon 28 Sep 2026, 1:30 pm.